

Scalability & Future Plan

Date	2 July 2026
Team ID	SWTID-2026-7714
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Trained on a limited dataset	May affect prediction accuracy on unseen data	High
2	Model performance may degrade over time	Accuracy can reduce as user patterns change	Medium
3	Currently a single model implementation	May not handle complex patterns as effectively	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for limited number of users	Deploy as a web application to handle multiple concurrent users
2	Data Storage	Dataset stored locally	Use cloud database (e.g., MySQL Cloud / Firebase) for scalable storage
3	Performance	Model inference on local machine	Optimize model and use cloud hosting for faster predictions
4	Security	Basic data handling	Implement user authentication and data encryption for secure access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
1	Add more features (e.g., spending behavior, payment history)	1 – 2 Months	Improve prediction accuracy
2	Integrate advanced ML algorithms	2 – 3 Months	Better model performance
3	Deploy as a web/mobile application	3 – 4 Months	Increased accessibility for users
4	Real-time model retraining with new data	4 – 6 Months	Up-to-date predictions
5	Add analytics dashboard for insights	6+ Months	Better decision support with insights